AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Employee Payroll Data Cleaning and Financial Analysis

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

Organizations across industries generate vast amounts of employee payroll data every month. However, this data is often raw, inconsistent, incomplete, or poorly structured, making it difficult to perform accurate financial analysis. Payroll records may contain duplicate entries, missing values, formatting errors, or misaligned structures, which hinder HR and finance teams from extracting meaningful insights.

Decision-makers rely on clean, accurate, and well-visualized payroll data to ensure compliance, manage costs, and make informed workforce-related decisions. Yet, without systematic data cleaning and analysis, payroll information remains underutilized and prone to errors that can affect both employees and organizational efficiency.

At the same time, the rapid growth of Artificial Intelligence and Generative AI tools such as Microsoft Copilot and AI-powered chatbots presents both opportunities and challenges. Professionals who can apply prompt engineering techniques to integrate AI into payroll workflows can significantly accelerate tasks like data validation, anomaly detection, and financial reporting. However, many users lack the skills to harness these tools effectively.

This project addresses these challenges by applying structured data cleaning, financial analysis, and visualization techniques to employee payroll datasets, while demonstrating how AI integration can enhance accuracy and efficiency. The solution aims to bridge the gap between raw payroll data and actionable insights, delivering value to HR teams, finance departments, and organizational leadership.

Key Points to Include:

Domain Background: Payroll management is critical for HR and finance, but raw data often contains inconsistencies.

Significance: Errors in payroll data can lead to compliance risks, financial misreporting, and employee dissatisfaction.

Gap Addressed: Lack of structured workflows and AI integration in payroll data handling.

Affected Stakeholders: HR managers, finance teams, employees, and organizational leaders.

Value Delivered: Clean, reliable payroll data that supports compliance, accurate financial analysis, and informed decision-making.

4. Objectives

- Collect and explore payroll data: Import the raw employee payroll dataset into Excel, examine its structure, fields, and identify data quality issues such as duplicates, missing values, or inconsistent formats.

- Clean and preprocess the dataset in Excel: Apply Excel-based techniques (sorting, filtering, conditional formatting, formulas) to remove duplicates, handle missing values, correct formatting errors, and standardize fields for consistency.

- Perform payroll-specific financial analysis: Use Excel formulas and pivot tables to calculate key payroll metrics such as gross salary, net salary, overtime payments, deductions, and department-level cost breakdowns.

- Develop structured reports and dashboards: Build clear, interactive Excel dashboards and pivot charts that visualize payroll trends, salary distributions, and workforce-related insights for HR and Finance stakeholders.

- Leverage AI-assisted techniques (optional): Explore how Microsoft Copilot or AI prompts can accelerate repetitive tasks such as formula generation, anomaly detection, or report summaries, complementing Excel workflows.

- Document and share the project process: Record the complete workflow, including dataset preparation, cleaning steps, analysis, and dashboard creation. Upload all files to GitHub and prepare a demonstration video showcasing the methodology and results.

- Ensure compliance and accuracy (domain-specific objective): Validate payroll calculations against standard rules (e.g., overtime rates, deductions, bonuses) to minimize errors and ensure financial accuracy.

- Support HR and Finance decision-making (domain-specific objective): Provide actionable insights that help HR teams manage compensation structures and finance departments optimize payroll expenses.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	➢ Obtained the raw payroll dataset in Excel format. ➢ Reviewed field names such as Employee ID, Department, Designation, Basic Pay, Allowances, Overtime, Bonuses, Deductions, Gross Salary, Net Salary, and Pay Month. ➢ Checked the dataset size and structure to understand the scope of records and identify potential data quality issues.
2	**Data Cleaning & Preprocessing**	➢ Used Excel functions and tools (Remove Duplicates, IF, IFERROR, Text-to-Columns, Conditional Formatting) to clean the dataset. ➢ Handled missing values, corrected formatting inconsistencies, and standardized text and date fields. ➢ Ensured salary-related fields (Basic Pay, Allowances, Overtime, Bonuses, Deductions) were properly aligned for accurate calculations.
3	**Analysis & AI Integration**	➢ Applied Excel formulas such as SUMIF, COUNTIF, VLOOKUP to calculate payroll metrics. ➢ Built pivot tables to analyze salary distributions, overtime costs, deductions, and department-wise payroll expenses.

		➢ Derived insights into employee compensation trends and financial allocations across HR, Finance, IT, and Sales departments.
4	**Visualization / Output**	➢ Designed interactive Excel dashboards with pivot charts, slicers, and KPI summaries. ➢ Visualized payroll trends such as gross vs. net salary comparisons, overtime contributions, and department-level salary breakdowns. ➢ Structured the outputs to be clear and actionable for HR and Finance stakeholders.
5	**Documentation & Submission**	➢ Documented the entire workflow, including dataset preparation, cleaning steps, analysis, and dashboard creation. ➢ Prepared the Excel file with all formulas, pivot tables, and dashboards. ➢ Uploaded the project files to GitHub and recorded a demonstration video explaining the methodology and results.

6. Resources and Timeline

A. Hardware & Software Requirements

- ***Laptop*** : **Intel(R) Core (TM) i5-2430M CPU @ 2.40GHz, 6.00 GB Installed RAM**
- ***Operating System:*** Windows 11 Pro 23H2
- ***Microsoft Excel :*** Excel 2016 version
- ***Microsoft Copilot :*** Available via Microsoft 365 subscription — for AI-assisted productivity
- ***AI Chatbot (Microsoft Copilot Web) :*** for prompt-based projects tasks and automation
- ***Internet Connection :*** Stable connectivity required for Copilot and AI chatbot access
- ***Web Browser :*** Latest version of Microsoft Edge

B. Dataset

- ***Availability:*** Clearly state whether the dataset is already prepared or needs to be sourced.

- ***Source:*** If publicly available, provide the source link (e.g., Kaggle, UCI Machine Learning Repository, Edunet-provided datasets).

- ***Collection/Generation***: If not available, explain how the dataset will be collected, generated, or simulated (e.g., synthetic payroll records created in Excel for HR/Finance analysis).

- *AI Prompt Projects:* For prompt-based projects, include a description of the use case along with reference content used as input.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

By the end of the project, the team aims to deliver the following outcomes:

- ***Clean and Validated Dataset:*** A thoroughly cleaned and validated payroll dataset, with all preprocessing steps documented. A clear before-and-after comparison will highlight improvements in data quality.

- ***Analytical Output:*** An interactive Excel dashboard or structured analytical report that effectively communicates key payroll and financial insights across departments, roles, and time periods.

- ***AI Integration:*** Demonstrated use of AI tools (Microsoft Copilot or AI Chatbot) to generate insights, summaries, or content outputs. All prompts and results will be documented to showcase practical AI applications in payroll analysis.

- ***Project Repository:*** A GitHub repository containing all project files, including the dataset, analysis scripts, dashboards, and a completed fellowship project report. A video demonstration will be provided to showcase the working output.

- ***Domain-Specific Deliverables:*** For this payroll-focused project, additional outcomes include a prompt library for HR/Finance analytics, a published Excel dashboard with salary trends and KPIs, and a structured framework for payroll data cleaning and financial analysis that can be reused in future projects.

8. References

The following resources, tutorials, and documentation were consulted during the project:

- ***Edunet Foundation :*** AICW Fellowship Program Guidelines, 2026. Available at: -- https://www.edunetfoundation.org
- LinkedIn Learning — AI and Data Analytics Courses, 2026. Available at: https://www.linkedin.com/learning

- ***Microsoft Corporation :*** Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

- ***Microsoft Corporation :*** Microsoft Copilot. Available at: https://copilot.microsoft.com

- ***Kaggle Inc. :*** Open Datasets. Available at: https://www.kaggle.com/datasets

- ***Project-Specific References:*** Additional materials such as research papers, GitHub repositories, YouTube tutorials, blogs, or other documentation consulted during the payroll dataset analysis and Excel dashboard creation.